

$$H_0: \mu \geq 100 \quad H_1: \mu < 100$$

$$\textcircled{2} \text{ given } \bar{x} = 80 \quad s = 20 \quad n = 100$$

$$\text{at } 95\% : 80 \pm 1.96 \cdot \frac{20}{\sqrt{100}}$$

$$97.5\% : 80 \pm 2.17 \cdot \frac{20}{\sqrt{100}}$$

$$\bar{x} = 34 \quad s = 12$$

$$\textcircled{a} P(28 < X < 31)$$

$$= \left(\frac{31-34}{12} \right) - \left(\frac{28-34}{12} \right)$$

$$= \Phi(-0.25) - \Phi(-0.5)$$

$$= 1 - \Phi(0.25) - (1 - \Phi(0.5))$$

$$= 1 - (0.5987) - (1 - 0.69146)$$

$$= 0.0934$$

$$\textcircled{b} P(X > 30) = 1 - P(X \leq 30)$$

$$= 1 - \left(\frac{30-34}{12} \right) = 1 - \Phi(0.33)$$

$$= 1 - (1 - 0.6293) = 0.6293$$

$$\textcircled{c} \text{ Median } = \text{Per } 50 = \Phi(0.00) = 0.5$$

$$0.00 = \frac{X-34}{12} \Rightarrow X-34 = 0$$

$$X = 34$$

$$\text{C.L.T } \textcircled{d} n=60 \Rightarrow P(X \leq 33) = P\left(Z \leq \frac{33-34}{\sqrt{60}}\right)$$

$$= 1 - \Phi\left(\frac{1}{\sqrt{60}}\right) = 1 - \Phi(0.129)$$

$$= 1 - 0.7421 = 0.2579$$

$$2. \bar{x} = 91.4193 \quad s = 12 \quad n = 10$$

$$\textcircled{a} P\text{-value} = P\left(T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}\right)$$

$$= \frac{91.4193 - 100}{12/\sqrt{10}} = -1.3567$$

$$d.f. = 9$$

$$P = 0.025$$

$$\textcircled{b} \text{ New } \mu = 5\% = 0.05$$

$$\therefore \text{reject } H_0$$

$$\textcircled{c} \text{ New } \mu = 1\% = 0.01$$

$$3. \text{ reject } H_0$$

$$O_i$$

$$E_i$$

$$310$$

$$350$$

$$265$$

$$250$$

$$115$$

$$100$$

$$310$$

$$300$$

$$(\text{table value})$$

$$\chi^2 = \frac{(O_i - E_i)^2}{E_i}$$

$$\chi^2 = \frac{(310-350)^2}{350} + \frac{(265-250)^2}{250} + \frac{(115-100)^2}{100} + \frac{(310-300)^2}{100}$$

$$= \frac{1600}{350} + \frac{225}{250} + \frac{225}{100} + \frac{100}{100} = 8.05476$$

$$\chi^2 \text{ table } \rightarrow 0.05 \chi^2 > 0.025$$

$$d.f. = (4-1) = 3$$

$$\chi^2 = 8.05476$$

$$\text{New } 0.05 \rightarrow \text{reject } H_0$$

$$0.01 \rightarrow \text{reject } H_0$$

$$\text{4. } \bar{x} = 80 \quad s = 20 \quad n = 100$$

$$\text{at } 95\% : 80 \pm 1.96 \cdot \frac{20}{\sqrt{100}} = (76.0800, 83.9200)$$

$$97.5\% : 80 \pm 2.17 \cdot \frac{20}{\sqrt{100}} = (75.6600, 84.3400)$$

$$5. \text{ New } \bar{x} = 64.4$$

$$\text{b) in } r = 1 - 0.9 \dots$$